

Technical Portfolio: Enterprise Architecture, Cloud Infrastructure & Systems Engineering

I. Executive Expertise Pillars

1. Enterprise Cloud Infrastructure & DevSecOps (Oracle / OCI)

- **Oracle Visual Builder Studio:** Engineered cloud-native, low-code/no-code development capabilities optimized for Oracle Cloud Infrastructure (OCI) and Oracle Cloud Applications.
- **Continuous Integration & Automated Testing:** Modernized the Oracle Developer Cloud Service pipeline by architecting automated build validation frameworks, including Hudson CI configuration jobs, end-to-end Selenium test suites, and QUnit validation tools.
- **Oracle JDeveloper / ADF Framework:** Served as a lead architect for core framework capabilities. Designed and implemented EJB/Bean Data Controls, design-time/runtime data binding, Sparse Beans, Bean Annotations, and reusable List of Values (LOV) components.

2. Database Utilities & Enterprise Lifecycle Tools (PeopleTools)

- **Change Assistant & Lifecycle Automation:** Designed the user interface, Wizard Framework, and security architecture for the PeopleSoft Change Assistant. Built the Download/Update Manager and environment data synchronization engines.
- **Data Mover & Enterprise Storage:** Authored the core data migration formats, high-performance indexing systems, and large-file transport mechanisms for Data Mover. Integrated the companion Process Scheduler utility.
- **Database Optimization & Tooling:** Engineered the PeopleTools Visual Compare Merge Tools, Enterprise Database Sizing utilities, Target Database setups, and executed deep SQL Performance Tuning across complex enterprise datasets.
- **Middleware & Connectivity:** Built secure proxy architectures for BEA Tuxedo, developed XML Web Services via Integration Broker, and engineered custom compiler/utility programs ([Swapbase](#) language tool, [Asciicheck](#)).

3. Hardware Engineering & Embedded Systems (Elonex)

- **Pentium Motherboard Design:** Formulated the layout, logic, and hardware architecture for high-performance Intel Pentium-based motherboards.
- **Network & Development Hardware:** Engineered an ASIC-level 10BaseT Ethernet card and designed an Integrated Development Toolkit hardware device.
- **Systems Architecture:** Deep expertise spanning Ram Disk Drives, MiniPC form factors, and early smartphone/mobile hardware topologies.

4. Cross-Industry Enterprise Software (ADP & Consulting)

- **Automotive Retail Solutions (ADP):** Spearheaded the Windows modernization strategy for ADP by developing the native Windows versions of *Shoplink* and *Penpro*, alongside the *Accupro* data conversion utility.

- **Industrial Analytics (Silicon Image Corp):** Programmed automated data analysis software to track, map, and isolate manufacturing defects on silicon wafers.
- **Enterprise E-Commerce & Logistics:**
 - Developed a large-scale data warehouse application for **Borders Books**.
 - Engineered the *Intrepid System* core merchandising application for **Montgomery Wards**.
- **Network Infrastructure (Capital Mitsubishi):** Architected and deployed the secure Local Area Network (LAN) infrastructure for a multi-site auto dealership.

II. Distinguished Projects & Research

- **Aerospace & Nanosatellites (NASA Ames / CREST):** Directed specialized engineering efforts for the *PharmaSat* satellite project, bridging complex software controls with space-bound telemetry.
- **Mobile & Ubiquitous Computing (Early Innovation):** Pioneered early-stage prototypes including a Bluetooth-enabled mobile shopping cart framework, clinical data gathering applications, and early Java Web Server Engines.

III. Technical Skills & Systems Inventory

Category	Technologies & Core Competencies
Cloud & Frameworks	Oracle Visual Builder Studio, OCI, JDeveloper/ADF, BEA Tuxedo, Integration Broker, Web Services
Data & DevOps	Data Mover, Change Assistant, Process Scheduler, Selenium, Hudson, QUnit, SQL Performance Tuning
Hardware & Systems	Pentium Architecture, 10BaseT Ethernet, LAN Topology, Embedded Systems, C/C++ Utilities